

User Manual

Setup Guide | Simulation | Operation

Varsha Vemula | AT89C51 Digital Clock & Stopwatch

1. Prerequisites

Software	Version	Purpose
Keil μ Vision	Version 5+	Compile firmware and generate HEX file
Proteus	Version 8 Pro	Load circuit and run simulation
Git	Any	Clone the repository
AT89C51 Pack	Keil Device DB	Required for MCU target in Keil

2. Getting the Code

Clone the repository from GitHub using the following command:

```
git clone https://github.com/varshavemula/digital-clock-stopwatch.git
cd digital-clock-stopwatch
```

3. Build Firmware in Keil μ Vision

Step	Action
1	Launch Keil μ Vision 5

2	Go to Project → Open Project and select the <code>.uvprojx</code> file from the <code>keil/</code> folder
3	Confirm the target MCU is set to AT89C51 under Project → Options for Target
4	Press F7 or go to Project → Build Target to compile the firmware
5	On successful build, a <code>clock.hex</code> file is generated inside <code>keil/output/</code>
6	Note the full path to <code>clock.hex</code> — it will be needed in Proteus

4. Run Simulation in Proteus 8

Step	Action
1	Launch Proteus 8 Professional
2	Open the simulation file: <code>proteus/simulation.pdsprj</code>
3	Double-click the AT89C51 component on the schematic
4	In the properties window, browse and load the <code>clock.hex</code> file from <code>keil/output/</code>
5	Set the crystal frequency to 11.0592 MHz in the same properties window
6	Click OK to save, then press the Play (■) button at the bottom to start simulation
7	The LCD should display the current time starting from 00:00:00

5. Operating the Device

Once the simulation is running, the LCD shows the current time in **HH:MM:SS** format. Use the three buttons to interact with the system:

Button	Pin	Clock Mode	Stopwatch Mode
BTN 1	P1.5	Switch to Stopwatch	Switch back to Clock
BTN 2	P1.6	No action	Start / Pause stopwatch
BTN 3	P1.7	No action	Reset stopwatch to 00:00:00

Note: A short buzzer beep confirms every button press. The clock continues running in the background even when the stopwatch mode is active.

6. Project Structure

Path	Description
src/main.c	Main firmware — clock logic, ISR, LCD driver
keil/project.uvprojx	Keil µVision project configuration file
keil/output/clock.hex	Compiled HEX binary for simulation/flashing
proteus/simulation.pdsprj	Proteus 8 circuit and simulation file
docs/Project_Report.pdf	Full project overview and objectives
docs/Hardware_Design.pdf	Circuit schematic, BOM, and pin mapping
docs/User_Manual.pdf	This document — setup and operation guide
README.md	GitHub landing page with project summary

7. Troubleshooting

Issue	Possible Cause	Fix
LCD shows no text	HEX not loaded in Proteus	Re-load clock.hex in AT89C51 properties
LCD shows garbled text	Wrong crystal frequency	Set oscillator to 11.0592 MHz
Buttons not responding	Simulation not running	Click Play (▶) in Proteus
Keil build fails	Missing AT89C51 device pack	Install pack via Pack Installer
Time not incrementing	Timer ISR not initialized	Check Timer 0 config in main.c